DFMEA - Long-Endurance Drone Cell (design R6-V9)

VBF-T8R1NOFORCE-DFMEA-01 | case t8_r1_noforce | 2026-08-25 | qualitative analysis - ratings are engineering judgement, not measurements

Scope: cell-level failure modes of the final virtual design (NMC811/graphite pouch, 65 x 1891 mm electrodes, 10 um separator, 6/8 um collectors, 4C fast charge at 45 C, 5C discharge). S/O/D 1-10 (10 worst); RPN = S x O x D.

#	Failure mode	Effect	S	Cause	O	Current controls	D	RPN	Recommended actions
1	Plating at charge end under cold conditions	dendrites, short risk, fade	7	anode-side transport polarization grows at low T (margin +6.3 mV at 45 C)	4	anode kinetics sized (r_n 1.2 um) + high-D electrolyte + h=60	3	84	BMS 45 C charge-temperature floor; 3C max below 25 C; re-simulate plating at 25 C
2	Separator (10 um) puncture / melt-shrink	internal short, thermal event	8	mechanical defect, dendrite (mode 1), winding tension	3	thin separator is ED trade; no coating in simulation	5	120	ceramic-coated separator qualification; supplier defect screening; abuse runs in next case
3	Cu foil (6 um) tear during winding	local overcurrent, hot spot	5	ultra-thin foil slitting/winding stress	4	design assumes intact collector	3	60	hardened Cu alloy foil; tension control on 1.89 m winding
4	Incomplete wetting of 1.891 m electrode	local starvation, early plating	5	long electrode path; 10 um separator restricts wicking	4	high D reduces transport consequence, not wetting	4	80	vacuum-fill with wetting hold; injection point design; formation capacity check
5	Pack cooling short of h=60 assumption	over-tempera ture at 5C / 4C	6	pack airflow design shortfall	2	h is design freedom; BMS cut-off not modeled	3	36	ducted prop-wash CFD at pack level; derate 5C above 30 C ambient
6	Current crowding on 1.89 m electrode	local heating, uneven aging	4	electrode length, single-tab assumption	4	uniform-current assumption (1D model limit)	3	48	multi-tab / full-width tab design study before build
7	Nano-NMC exothermic release at elevated T	thermal runaway in abuse	6	high specific surface of nano-particles	2	thermal red line 333.15 K enforced (T_max 326.8 K simulated)	5	60	run overcharge/nail protocols in safety follow-up case

Top risks: #2 separator (RPN 120), #1 plating margin (RPN 84), #4 wetting (RPN 80). None are criterion failures in the entry-0 sense - they are manufacturing/operational risks of an aggressive virtual design, flagged for the next iteration (physical prototype case).